

Zomato Restaurant Data Analysis

A Data Analytics & Business Intelligence Project

Python • Pandas • NumPy • SQL • Power BI • Data Visualization

Project Type	Data Analytics Project
Dataset	Zomato Restaurant Dataset
Records Analyzed	780

1. Executive Summary

This project analyzes 780 restaurant records to understand restaurant distribution, customer ratings, votes, pricing, online ordering and table booking behavior. The overall average rating is **3.69/5**, while the average cost for two people is approximately **₹458**.

Dining restaurants account for approximately **70.0%** of the dataset.

2. Introduction

The restaurant industry generates large amounts of customer and business data.

3. Problem Statement

Restaurant businesses need to understand what factors are associated with customer ratings and popularity.

4. Project Objectives

- Understand the structure and characteristics of the dataset.
- Clean and preprocess restaurant data.
- Analyze restaurant categories and their distribution.
- Study customer rating patterns.
- Identify actionable business insights.
- Create visualizations and a Power BI dashboard.
- Prepare a professional analytical report.

5. Technology Stack

Technology	Purpose
Python	Data analysis and automation
Pandas	Data manipulation and cleaning
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical visualization
SQL	Business queries and analysis
Power BI	Interactive dashboard
ReportLab	Automated PDF report generation

6. Dataset Description

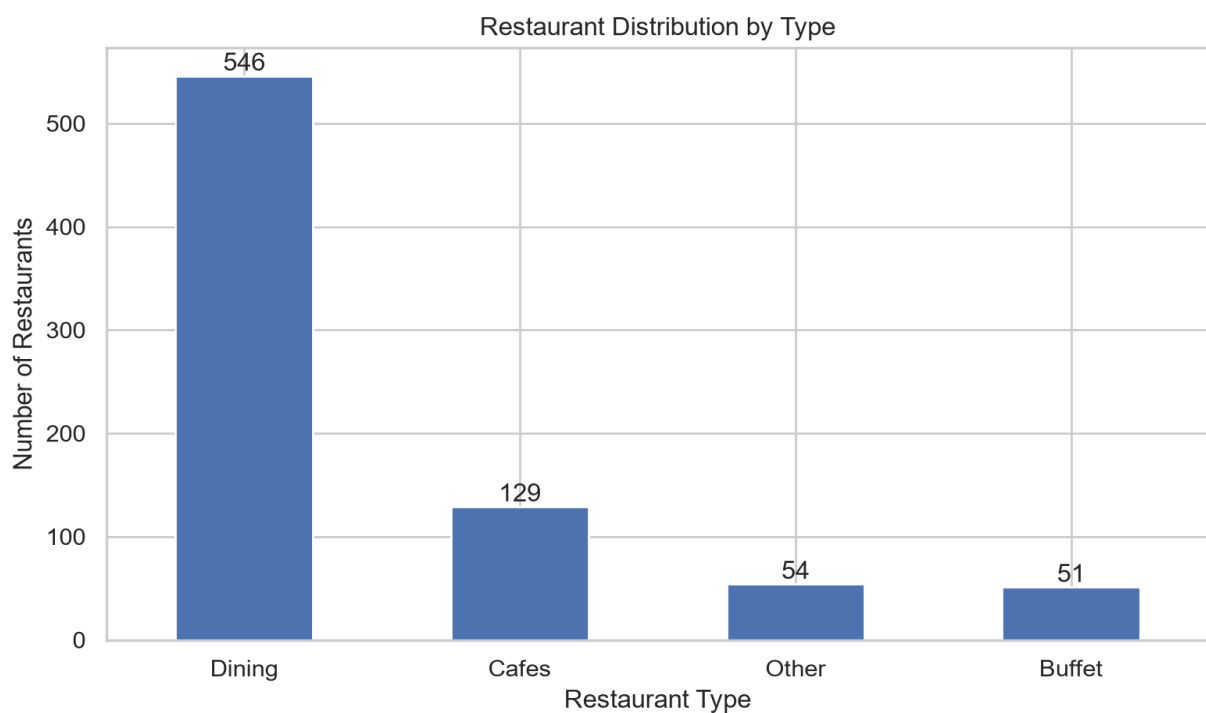
Metric	Value
Original Records	1,000
Final Records	780
Columns	7

7. Data Cleaning & Preprocessing

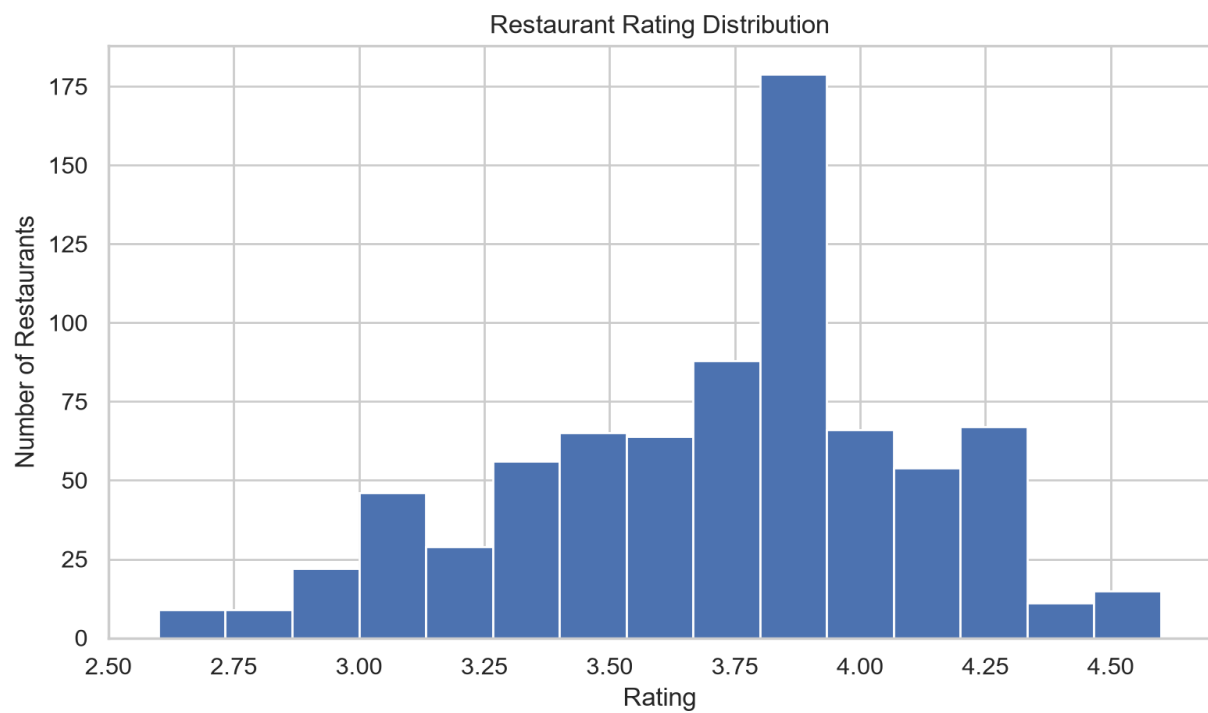
- Removed duplicate restaurant records.
- Removed '5' from rating column.
- Converted numeric fields.
- Standardized categorical values.

8. Exploratory Data Analysis

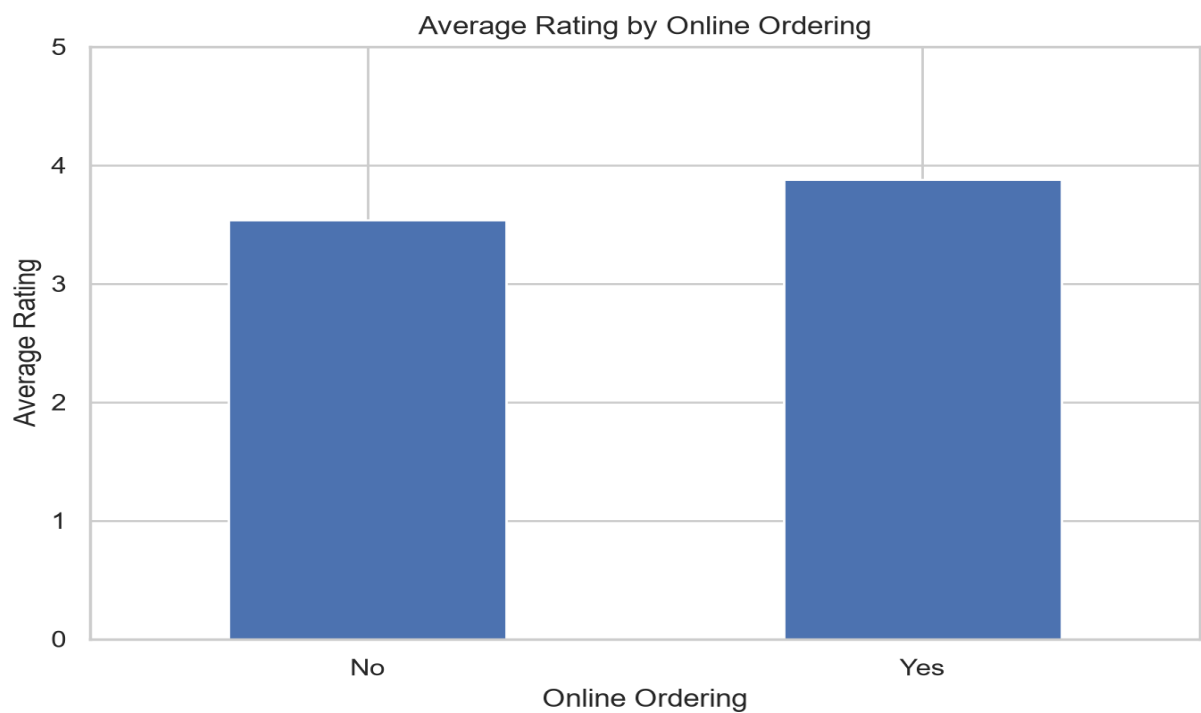
8.1 Restaurant Distribution



8.2 Rating Distribution



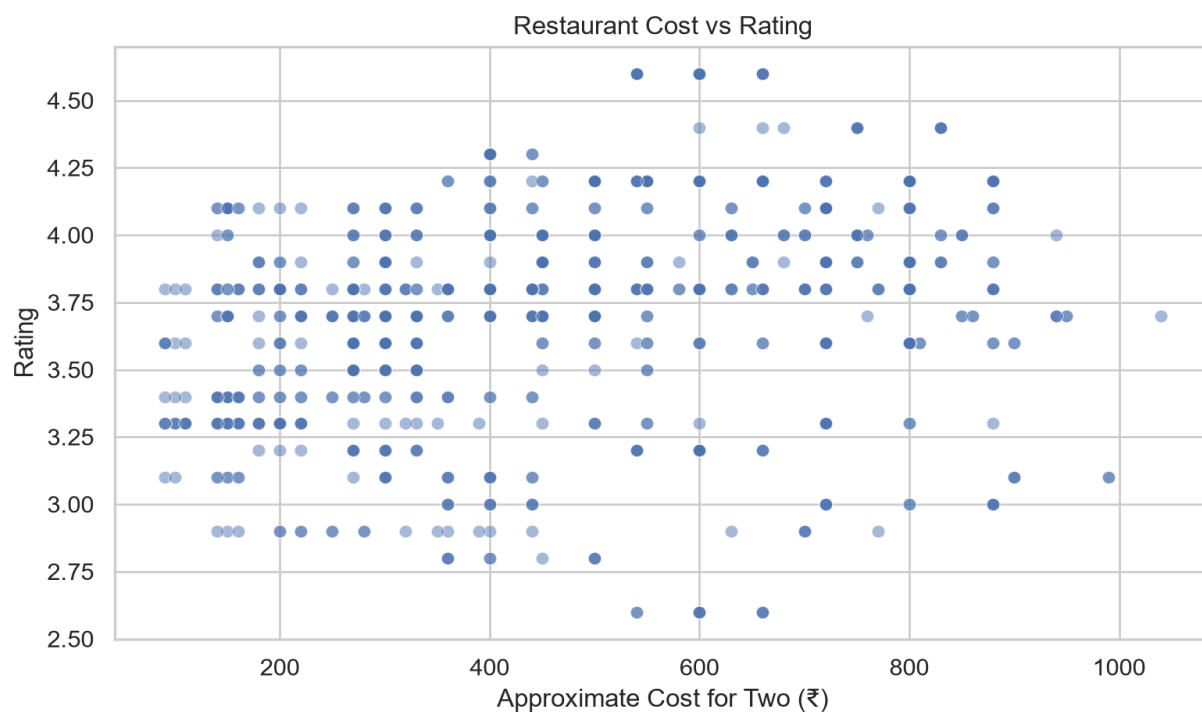
8.3 Online Ordering and Rating



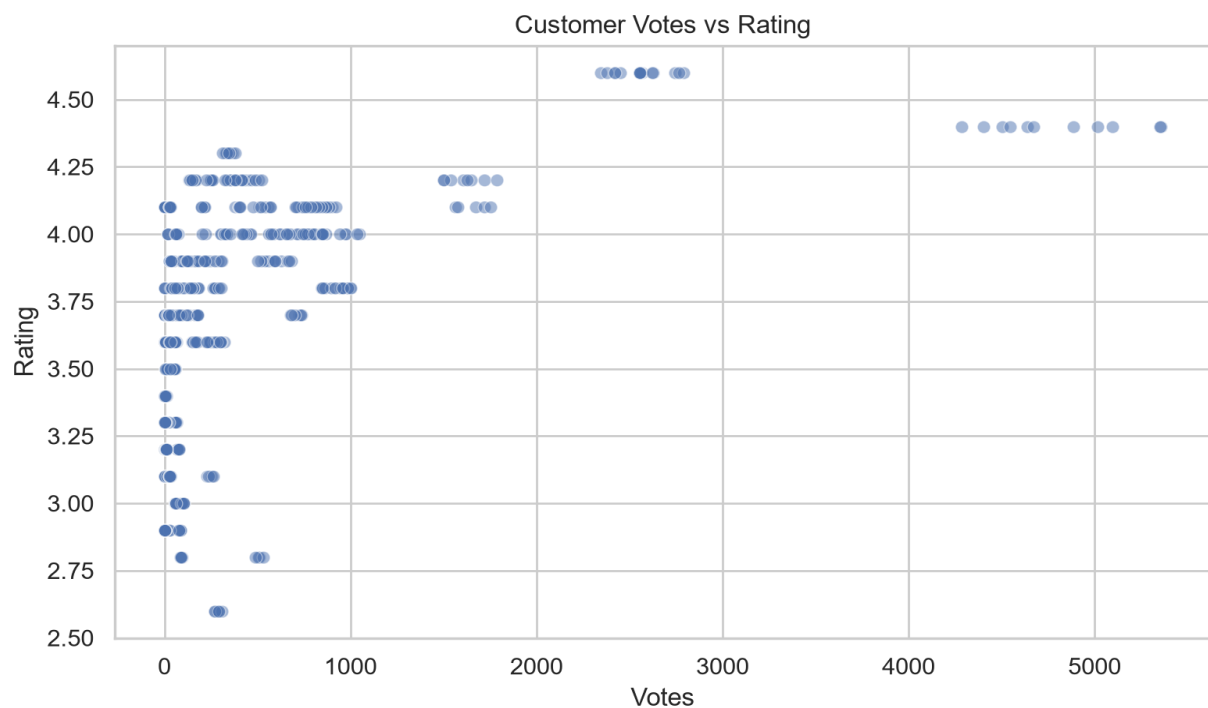
8.4 Table Booking and Rating



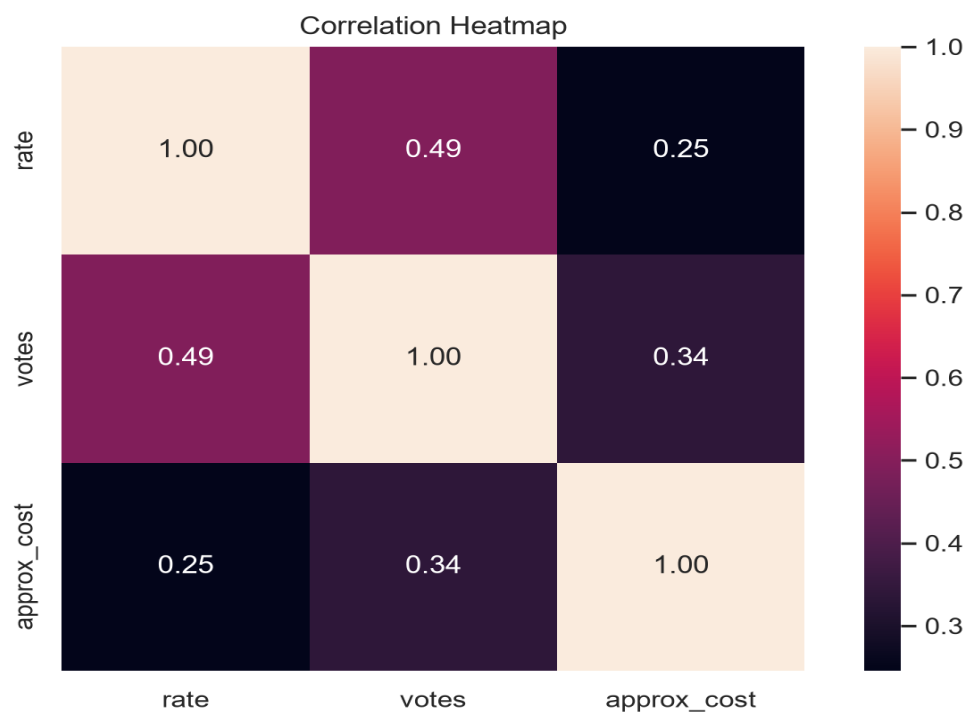
8.5 Cost vs Rating



8.6 Votes vs Rating



8.7 Correlation Heatmap



9. Business Analysis

KPI	Value
Total Restaurants	780
Average Rating	3.69 / 5

KPI	Value
Total Votes	253,772
Average Cost	■458

10. Business Recommendations

1. Introduce Online Ordering
2. Offer Table Reservations
3. Focus on Customer Experience
4. Maintain Competitive Pricing